Computers & Industrial Engineering 194 (2024) 110405

Contents lists available at ScienceDirect

Computers & Industrial Engineering

journal homepage: www.elsevier.com/locate/caie

Integrating discrete-event simulation and artificial intelligence for shortening bed waiting times in hospitalization departments during respiratory disease seasons

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ARTICLE INFO

Keywords:
Discrete-Event Simulation (DES)
Artificial Intelligence (AI)
Random Forest (RF)
Hospitalization Departments (HDs)
Seasonal Respiratory Diseases (SRDs)
Bed Waiting Time

ABSTRACT

Seasonal Respiratory Diseases (SRDs) usually produce a heightened number of Emergency Department (ED) attendances due to their rapid dissemination within the community and the ineffective prevention measures. Such a context requires effective management of the emergency care processes to provide in-time diagnosis and treatment to infected patients. Nonetheless, EDs have evidenced severe operational deficiencies during these periods, thereby provoking extended bed waiting times in Hospitalization Departments (HDs). Therefore, this paper presents a hybrid approach merging Artificial Intelligence (AI) and Discrete-Event Simulation (DES) to shorten the bed waiting times in HDs considering patient records collated in the first emergency care stages. First, we implemented Random Forest (RF) to estimate the probability of respiratory worsening based on socio-demographic and clinical patient data. Second, we inserted these probabilities into a DES model mimicking the emergency care from the admission to the HD. We then pretested different HD configurations and strategies seeking to reduce the HD bed waiting time. A case study of a European hospital group was used to validate the suggested framework. The AI-DES model enabled decision-makers to identify an improvement proposal with hospitalization bed waiting time lessening, oscillating between 7.93 and 7.98 h.

1. Introduction

With resources scarce worldwide and hospitals overcrowded, managing hospital admissions is becoming increasingly critical (Nan et al., 2024). Therefore, organizations need innovative methods and tools, such as Artificial Intelligence (AI), to handle the increase in demand and the paucity of resources (Alowais et al., 2023). Indeed, AI is now an integral part of demand planning for patient systems, and its use supports organizations in better dealing with challenging circumstances (Ali et al., 2023). It has been adopted worldwide, and most recently, the implementation of AI has also been experimented with to evaluate the likelihood of patient admission in the UK healthcare system (NHS England, 2023).

However, despite the increasing use of AI for supporting demand planning in healthcare contexts, there is still a gap in developing hybrid simulation approaches combining AI methods with simulation methods to manage the whole flow of demand planning starting from admission to the Emergency Department (ED) to the hospitalization of the patients. Those phases are critical, especially during the peak of respiratory diseases, and their management can be highly challenging (Arias et al.,

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https://doi.org/10.1016/j.cie.2024.110405

Available online 25 July 2024

2023). Indeed, worldwide, winter months are becoming a cause of concern in hospitals, with people struggling to find a bed and not being given the correct treatment. The need for more personnel in healthcare structures exacerbates this. The situation is even more critical given that doctors no longer prefer working in the ED anymore because of the harsh conditions (Pourmand et al., 2023).

Indeed, healthcare organizations need not only to estimate the number of patients admitted to the hospital but also to predict how the number of patients will evolve once admitted to the hospital in a more severe. At the same time, hospitals still have to provide healthcare services for all the patients, i.e., managing the capacity of the hospitals and patients not affected by respiratory diseases. Bed waiting time is critical when there is huge demand. Therefore, innovative methods must be employed to consider all the aspects listed above and create different scenario simulations to help managers allocate the correct number of resources for ED and bed capacity (Arora et al., 2023).

This paper presents a hybrid simulation combining AI techniques with the Discrete Event Simulation (DES) methods. First, we select which features should be chosen to estimate the likelihood of worsening respiratory disease requiring hospitalization. We take the features from an extensive analysis of the literature review. Once the choice of the predictors has been confirmed and its statistical significance has been tested, a dataset of data for patients has been coherently constructed thanks to an appropriate preprocessing phase that included removing patients where data were not consistent or not wholly available. Second, based on accuracy, a Random Forest (RF) model was selected as an AI algorithm. After the appropriate phase of testing and training, the RF allows for estimating the probability of patients being hospitalized. Third, we inserted these probabilities into a DES model simulating the emergency care required by the patients from admission to the ED until eventual hospitalization. The combination of those methods has the intention of reaching the following aims:

- The preprocessing phase proposes a robustly based selection of features to classify patients and allows for a reliable hybrid simulation approach.
- The RF method permits accurate prediction of the likelihood of requiring hospitalization, and it is beneficial when applied to large databases.
- The DES model allows for simulating the whole flow of patients and differentiating their potential outcomes. Its benefits relate to modeling diverse scenarios under very uncertain conditions, such as during a respiratory disease peak.

We applied the framework to a case study in a European hospital chain with many ED admissions.

To the best of our knowledge, this study depicts a novel hybrid simulation approach that deals with the evolution of the gravity of respiratory diseases that has yet to be reported in the literature. The study illustrates the critical role that AI and DES may play in designing in-time interventions that diminish the bed waiting times for incoming patients. The paper answers to the following research questions:

- Can hybrid simulation approaches describe the evolution of respiratory disease to model the demand of hospitalizing patients in overcrowded health environments?
- Can AI and DES techniques be merged to simulate bed waiting time in health systems?

Therefore, the objectives of the paper can be stated as follows:

- Developing an AI and simulation-based approach to predict the likelihood of the patients requiring severe treatments due to severe respiratory diseases;
- Simulating the demand capacity of health systems and management, i.e., the inflow of patients in the ED, their waiting time for a bed, and better management of the available resources.

The rest of the paper is structured as follows: Section 2 presents a literature review on AI and DES applications in real-world healthcare systems, whereas Section 3 details the proposed methodology. Section 4 illustrates the application of the suggested approach in a real case study, and Section 5 discusses the results. Finally, Section 6 outlines the practical and managerial implications of our approach whilst Section 7 depicts the conclusions.

2. Literature review

The simulation study encompasses modeling intricacies for understanding system complexities, simulation algorithms for system evolution, validation techniques for accuracy comparison, and optimization strategies for efficiency improvement (Padovano, et al., 2024). The Discrete-Event Simulation (DES) is applied to replicate the behavior of real-world systems when specific events occur in particular instances, offering valuable insights into system behavior and dynamics. The applications of such simulation have been explored across various fields, including manufacturing (Kampa, et al., 2017) (Greasley, 2005), transportation (Turner, et al., 2019), telecommunication (Tavanpour et al., 2022) and healthcare (Poursoltan, et al., 2022). In manufacturing sectors, DES is extensively applied for tasks such as inventory management (Bhosekar, et al., 2023), facility layout design (Zúñiga, et al., 2020), and optimizing changeable production systems (Albrecht, et al., 2014). Numerous studies have demonstrated the effectiveness of simulating complex systems in identifying bottlenecks, managing resources, and minimizing cycle times, among other benefits (Prajpat & Tiwari, 2017). In transportation and logistics, DES facilitates to optimize and improve performance of supply chain operations (Abideen & Mohamad, 2021), evaluate logistic systems (Hu, et al., 2023), and vehicle routing (Keskin, et al., 2021). Moreover, a study used DES for operational and maintenance procedures, addressing system failures arising from human error or spare parts unavailability, aiming to ensure uninterrupted telecom operations (Lyubchenko, et al., 2020). Similarly, research has been conducted on data analytics for optimizing call center operations (Serper, et al., 2022), and wireless networking (Tavanpour, et al., 2022). Moreover, DES has been employed in various studies within the healthcare sector to model patient flows within hospitals (Basaglia, et al., 2022), and design emergency departments (Morgareidge, et al., 2014), thereby aiding in the optimization of healthcare operations.

Nowadays, researchers are directing their attention to integrating DES with other techniques such as machine learning, data mining, data farming, process mining, and other big data analytics approaches to address contemporary challenges (Greasley & Edwards, 2021; Ortiz-Barrios et al., 2024). For instance, a study performed in the healthcare sector integrated DES with machine learning algorithms to analyze patient waiting times and the number of patients treated, considering input variables such as numbers of physicians, nurses, clerks, beds, and triage areas (Atalan, et al., 2022). Similarly, another research utilizing production data from manufacturing process demonstrated that DES is highly effective in generating the quality data essential for training machine learning models (Chan, et al., 2022). Moreover, a study addressed the shortcomings of existing simulation models and coupled DES with data mining techniques to pinpoint bottlenecks between emergency departments and hospital wards (Ceglowski, et al., 2016). Additionally, DES in conjunction with process mining and optimization techniques, is also utilized to address the issue of long queues in hospital emergency departments (Antunes, et al., 2019). Similarly, another study coupled DES with optimization techniques to analyze patient flow and allocate resources, aiming to improve healthcare delivery services (Jacobson, et al., 2013). Several additional research studies have explored the integration of DES and virtual reality to address complex data analysis problems and to establish connections of DES to real-time Big Data streams, aiming to fulfill the visualization needs of Industry 4.0 (Turner, et al., 2016).

Recently, in response to the challenges faced by Intensive Care Units (ICUs) during the COVID-19 pandemic, researchers coupled Artificial Intelligence (AI) with DES to effectively address issues such as the efficient management of bed capacity utilizing predictive modeling (Ortiz-Barrios, et al., 2023). Furthermore, another research focusing on manufacturing systems indicates that DES plays a crucial role in testing and training AI systems, while AI is instrumental in controlling, optimizing, and analyzing DES models (Kranz, et al., 2023). Similarly, another study integrating DES and artificial intelligence (neural networks) revealed that while artificial intelligence delivers superior qualitative estimates, DES excels in providing timely and efficient results and is powerful computational method for decision making (Wilson, et al., 2022).

Considering the potential of DES coupled with emerging technology, this study aims to address the challenge of managing patients during SRDs. A critical aspect of such emergencies is reducing patient waiting times. Thus, this research integrates DES and artificial intelligence to minimize bed waiting times in HDs and effectively manage these unpredictable situations.

3. Proposed methodology

The burden posed by SRDs entails the use of powerful methods laying the groundwork for designing remedies granting an upgraded response of the hospitalization departments against the forecasted demand. This response is partially supported by timely bed allocation for SRD patients evidencing high probability of health worsening if not intervened promptly in an upstream service. A hybrid approach integrating AI and DES (Fig. 1) has been proposed to cope with the hospitalization bed waiting time problem experienced by the SRD patients. On the one hand, AI will be used to predict the hospitalization admission probability based on sociodemographic and clinical data of SRD patients. These predictions will be later inserted into a simulation model mimicking the SRD patient journey within the medical center. The aim oi this model is two-fold: i) to evaluate whether the hospitalization department can provide timely bed allocation to SRD patients and ii) to pretest improvement scenarios aimed at diminishing the hospitalization bed waiting time. A detailed description of the proposed approach is provided below:

Phase 1 – Definition of candidate hospitalization admission predictors in SRD patients: Features potentially correlated with the hospitalization admission likelihood are extracted from the reported literature and SRD hospital patient records. Thus, we can determine which features are feasible to include in the dataset.

Phase 2 – Dataset preparation: The data corresponding to the predefined features are gathered from the hospital's medical records. We will then clean the data and address any erroneous registrations to avoid bias that may downgrade the performance of the AI model. Following this, the records are tagged, and balance between the classes is evaluated. In case of imbalance problems, random under-sampling on the training cohort should stabilize both layers.

Phase 3 – Significance assessment of potential predictors: Descriptive statistics are used to identify patterns that can explain the health evolution in hospitalized and non-hospitalized SRD patients. We then conduct comparative assessment and perform Analysis of Variance (ANOVA) tests ($\alpha = 0.05$) are performed to verify the statistical significance of each feature. If the p-value is less than α, the feature is categorized as predictor and should be incorporated into the AI model; otherwise, it is concluded as a non-significant factor and not hence useful for the hospitalization admission prediction. The Mean Decrease of Ginni Coefficient (MDGC) is also computed to determine the significance strength of the features.

Phase 4 – Training, testing, and validation of the AI model: The dataset is randomly partitioned in training and test cohorts using a 70/30 ratio. The Rstudio software is then utilized to create a Random Forest (RF) algorithm that will predict the hospitalization admission likelihood in SRD patients. Then, a 10-fold cross-validation is executed to adjust the model hyperparameters. Several performance metrics including sensitivity, specificity, accuracy, positive/negative predictive value, and the Area under Curve (AUC) are calculated to measure the discrimination ability of the AI model from different operational perspectives. Ultimately, the McNemar test ($\alpha = 0.05$) is applied to appraise the homogeneity assumption between the percentages of misclassified SRD patients. This hypothesis is confirmed if the resulting *p*-value is higher than the error reference.

Phase 5 – Simulation model diagramming: Direct observation and healthcare procedures are the main sources supporting the description of the SRD patient journey within the hospital. After close validation with the medical and administrative staff of the medical center, this journey is depicted through a flow diagram denoting what happens in the daily routine of the ED-ICU pathway. In this phase, process variables and parameters are also identified to describe this journey more detailedly.

Phase 6 – Input data analysis: Process variables data are collated according to the calculated sample size. These data are then preprocessed for randomness, homogeneity, and goodness of fit.

Phase 7 – Creation and validation of the DES model: The flow diagram and input data analysis underpin the creation of a virtual model mimicking the SRD patient journey. The model is profiled in simulation software and then validated to ensure comparability with the real-world system. Once the equivalence between the models has been achieved, performance diagnosis is performed to determine how well the medical center complies with the expected hospitalization bed waiting time.

Phase 8 – Pretesting improvement proposals: If the bed waiting time target is not satisfied, the healthcare managers are called to design improvement scenarios reducing the gap between the current and desired performance. Once this has been defined and iterated with the medical staff and operational management experts, the modelers create the scenarios in the model and pretest their performance to determine if they would be effective in the real context before implementation.

3.1. *Random forest*

Healthcare dynamics is complex and, therefore, full of variables and interactions that are difficult to model following the typical patterns of a dynamic system. Artificial Intelligence (AI) can address this methodological need and has gained increasing attention from the healthcare community. AI techniques have been shaping the landscape of healthcare operations several years ago, including interventions in clinical practice, medical diagnosis, treatments, and patient monitoring.

One of the most popular AI techniques is Random Forest (RF), which is an algorithm that is inserted in the Machine Learning (ML) branch. RF has been used in several applications comprising prediction, classification, and feature selection (Kaur et al., 2019; Jackins et al., 2021; Xue et al., 2022). This non-parametric technique employs multiple combinations of independent feature trees to forecast the outcome metric. The use of RF is supported by the following considerations: i) RF can be implemented in a two-class problem as the one defined with the hospitalization admission probability (transferred/non-transferred) (Utkin et al., 2020; Xu et al., 2020) ii) RF captures non-linear complex relationships and interrelationships between the covariates as the ones commonly found in healthcare implementations (Hong and Lynn, 2020; Holodinsky et al., 2021), iii) RF is vastly resistant to overfitting problems (Breiman, 2001) and can deal with a mixture of quantitative and categorical variables as found in the majority of health evolution prediction models (Casanova et al., 2014) iv) RF is highly accurate when predicting probabilities which is intended regarding hospitalization admission (Fernández-Delgado et al., 2014; Loef et al., 2022), v) RF can accurately cope with large high-dimensional datasets (Ghosh and Cabrera, 2021;

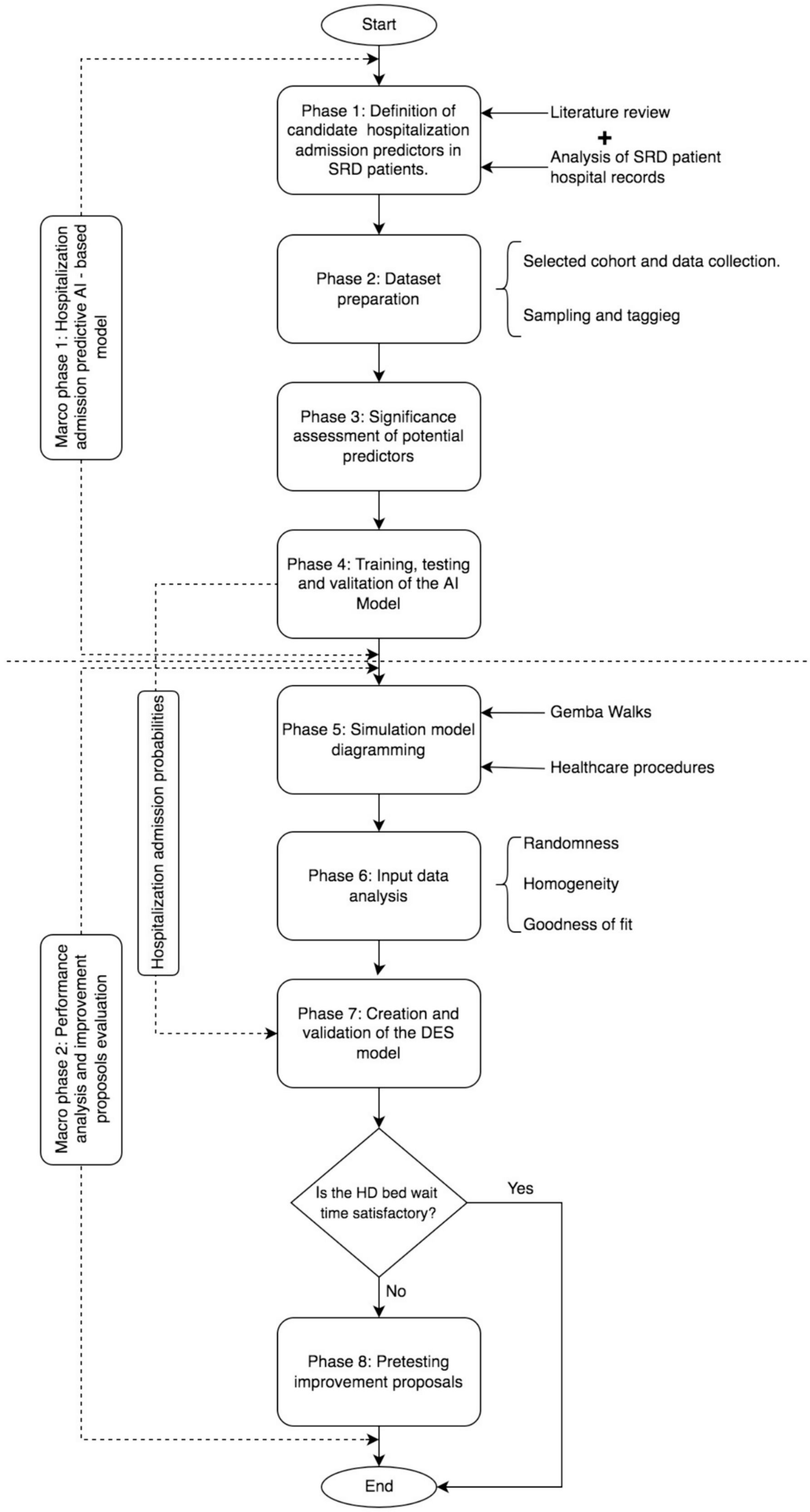

Fig. 1. The proposed AI-DES approach for lessening hospitalization bed waiting time.

Albuquerque et al., 2022), and vi) it is easy to comprehend by healthcare professionals who are not usually skilled in data-driven approaches (Loef et al., 2022).

RF employs bagging and feature randomness to derive an independent forest of decision trees. In other words, RF produces a random subgroup of predictors with very low correlation among decision trees. The diverse decision trees or sets of classifiers are denoted as $h(X, \ominus)$. Each tree is compounded by a data sample extracted from the training subset symbolized as vectors $< X, Y >$. One third of the training subset is usually employed as test data. Thereafter, the individual decision trees are averaged to generate the prediction (Fig. 2). During its functionality, three principal hyperparameters (node size, amount of trees, and number of sampled predictors) need to avoid potential bias in the model (Lujan-Moreno et al., 2018).

The mathematical formula supporting RF is as follows (Breiman, 2001):

The margin function of RF $mr(X, Y)$ is defined as stated in Eq. 1:

$$mr(X, Y) = P_{\ominus}(h(X, \ominus) = Y) - \max_{j \neq Y} P_{\ominus}(h(X, \ominus) = j) \tag{1}$$

On the other hand, the generalization error PE^{*} is defined as follows (Eq. 2):

$$PE^{*} = P_{X,Y}(mg(X, Y) < 0) \tag{2}$$

An important property of RF implies that PE* converges to $P_{X,Y}(P_{\ominus}(h(X, \ominus) = Y) - \max_{j \neq Y} P_{\ominus}(h(X, \ominus) = j) < 0)$ as the number of trees augments. This supports why RF does not overfit in presence of a higher amount of trees, which aggregates dynamism to the model without introducing bias. More details about the RF pseudocode can be consulted in Jackins et al. (2021).

3.2. Discrete-Event simulation (DES)

Understanding the interactions between processes is pivotal to managing the patient flow in hospitalization departments effectively. Inefficient internal supplier-customer relationships have been detected as contributors to poor health outcomes in hospitalized patients (Le lay et al., 2023). Therefore, it is necessary to count on a robust method providing an overall representation of the SRD patient journey, including the multiple treatment pathways and the transitory/absorbing states that may be derived in the function of the patient's health evolution (Gillespie et al., 2016). Thereby, the hospital administrators can detect bottlenecks, inefficiencies, non-value-added activities, and underused resources (Melman et al., 2021). DES can underpin these tasks given the following considerations:

i) DES describes the operational landscape of healthcare services by portraying their process-to-process interactions (McClean et al., 2011; Ortíz-Barrios et al., 2017),

ii) DES is useful for pretesting improvement strategies, avoiding the financial and operational affectations caused by the trial-and-error

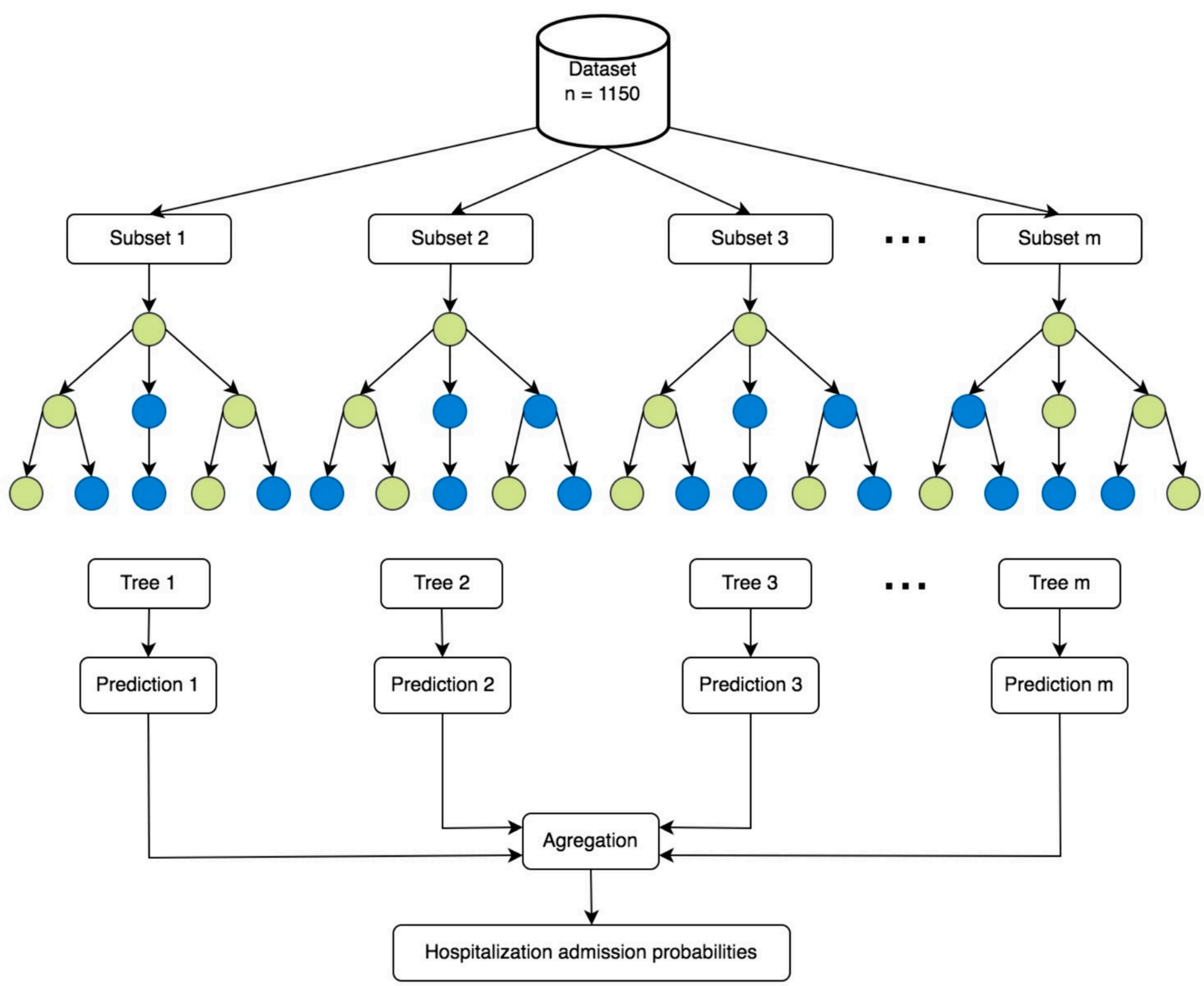

Fig. 2. Random Forest structure and functionality in the case of hospitalization admission prediction.

approach (Troncoso-Palacio et al., 2018; Ordu et al., 2021), iii) DES facilitates engagement with stakeholders usually by animating the entire SRD patient journey within the hospital,

iv) DES is useful for denoting time-dependent variables as the ones expected in the healthcare context (Nuñez-Pérez et al., 2017), and

v) DES offers a comprehensive set of performance metrics that support profound process analysis (Gul et al., 2016).

The step-by-step DES procedure commences with identifying the phases compounding the SRD patient journey within the medical center. In this respect, Gemba walks are conducted to validate and complement the information provided in the associated healthcare procedures. Another important outcome is the recognition of process parameters and variables. Thereafter, data collection plans are deployed to obtain the required sample to describe each variable. Then, autocorrelation tests are conducted to verify the randomness assumption of these variables. This hypothesis is accepted if the resulting T statistic oscillates between −2 and 2, which entails the definition of a stochastic distribution. Otherwise, ML techniques need to be used to predict the variable. Following this, homogeneity tests are performed to define whether there are subgroups of data within the primary dataset. In this regard, Welch's tests (error level = 0.01) are deployed to accept/reject the homogeneity. If the *p*-value is higher than the significance level, the data is categorized as homogeneous and only one probabilistic model is necessary to describe the variable; otherwise, one stochastic expression must be defined for each layer. Following this, Kolmogorov-Smirnov (KS) tests are undertaken to test the data's goodness of fit to different probabilistic distributions. *P*-values greater than the alpha level indicate that the data fits the proposed stochastic model. The next step entails the patient journey modelling through simulation software where input data analysis results and process flow diagrams converge to animate the system dynamics. The derived model should be validated through an equivalence analysis (Ortiz-Barrios and Alfaro-Saiz, 2020). To obtain the validation sample, it is necessary to initially iterate the model 10 times. Thereby, we can estimate the stability of the process variable and calculate the number of runs required for fully describing its behavior. If the *p*-value emanated from the 1 sample t-tests ($\alpha = 0.01$) is higher than this threshold, the model is concluded to be a good imitation of the real-world system; on the contrary, the modelers need to make sequential adjustments until achieving the desired statistical equivalence. In case of operational deficiencies, improvement scenarios may be proposed by the healthcare directors and medical staff. In this case, modelers are called to recreate this remedy in the simulated model, collect the hospitalization bed waiting time data, and carry out Mann-Whitney tests ($\alpha = 0.01$). If the *p*-value is less than the significance level, we can conclude that the proposed scenario is effective and would trigger a significant reduction in the hospitalization bed waiting time (W=105,618, *p*-value = 0.004) if implemented in the wild. Otherwise, the proposal must be discarded, and the management team should explore other alternatives.

4. Results

Administering bed-installed capacity in HDs may be pivotal in defining the future pathway of patients with SRD. In other words, prolonged waiting periods for HD beds may delay effective treatment and health monitoring, provoking an increased likelihood of respiratory failures and the associated elevated mortality rates and service costs. In line with this, a European medical center moved towards developing a robust predictive decision-making framework assisting HD managers in creating preventive operational measures that accelerate access to beds when needed. The healthcare group has invested in standardizing the measure culture by implementing training programs supported by pertinent data management software. Therefore, this organization has constructed valuable datasets that can be employed for feeding and training AI-DES models, offering meaningful insights to the HD operational planning. The datasets may provide an overview of the SRDs impact on HD performance and, therefore, lay the groundwork for extracting patterns of different process variables related to the hospitalization bed waiting time. Formal consent was provided by the ethics committee of the medical center (Agreement number: 14–12-2021–004; Access request ID:39) to use the data in a project aiming at shortening the bed waiting times experienced by infected people by using an integrated AI-DES approach. The subsequent sub-sections will present the results derived from the proposed methodology specifying how these insights are interpreted and then utilized by the decision-makers to delineate the new HD configurations addressing the expected SRDs.

4.1. Candidate predictors of hospitalization in respiratory-affected patients

The treatment options and health states characterizing SRD patients may be delineated by the influence of diverse sociodemographic and clinical factors collated during the healthcare service. Nonetheless, there is a need to anticipate using scarce resources that are vital for ensuring the most effective care in the presence of demand peaks. It is then essential to identify the hospitalization drivers based on models using significant variables gathered from the downstream stages of the healthcare journey. Table 1 depicts the features hypothesized by the reported literature and the medical staff as contributors to hospitalization admission in SRD patients. These characteristics are included in the medical center's raw dataset, and testing their statistical significance on the hospitalization admission probability is feasible.

4.2. Dataset preparation

4.2.1. Selected cohort and data collection

The raw dataset supplied by the hospital group encompasses the anonymized clinical records of 4479 SRD patients aged between 1 and

Table 1
List of hospitalization admission features in SRD patients.

Feature	Feature description	Previous studies using the feature
Sex (S)	Sex of the SRD patient.	Silveyra et al. (2021); Patel et al. (2021); Chiu et al. (2022)
Age (A)	Age of the SRD patient.	Raita et al. (2020); Kang et al. (2021); Wu et al. (2021)
Diastolic Arterial Pressure (DAP) (2 records: DAP_1, DAP_2)	Diastolic arterial pressure measured at the triage station (DAP_1) and four hours after admission (DAP_2).	Goto et al. (2018); Peng et al. (2020); Aktar et al. (2021)
Systolic Arterial Pressure (SAP) (2 records: SAP_1, SAP_2)	Systolic arterial pressure measured at the triage station (SAP_1) and four hours after admission (SAP_2).	
Heart Pulse (HP) (2 records: HP_1, HP_2)	Heart pulse (beats per minute) measured at the triage station (HP_1) and four hours after admission (HP_2).	Bolourani et al. (2021); Kumar et al. (2022); Ortiz-Barrios et al. (2023)
Oxygen Saturation Level (OSL) (2 records: OSL_1, OSL_2)	Oxygen saturation level measured at the triage station (OSL_1) and four hours after admission (OSL_2).	Le et al. (2020); Raita et al. (2020); Patel et al. (2021)
D-Dimer Concentration (DDC)	The d-Dimer level detected by the emergency care lab after four hours of admission.	Ferrari et al. (2020); Patel et al. (2021); Gawlitza et al. (2021)
Lactate Dehydrogenase (LDH)	The Lactate Dehydrogenase concentration (UI/L) calculated by the emergency care lab after four hours of admission.	De Fátima Cobre et al. (2021); Liu et al. (2022); Iwase et al. (2022)
C-Reactive Protein Level (C-RPL)	The C-reactive protein concentration (mg/ml) calculated by the emergency care lab after four hours of admission.	Peng et al. (2020); Stokes et al. (2021); Tan et al. (2021)

100 years. The specific patient data was obtained from the patient data management software and comprised information on vital signals, lab results, hospitalization admission, sociodemographic profile, and outcomes. These people were served at the medical center ED between February 2020 and February 2021. The SRD confirmation was obtained after triage and validated through blood test results.

4.2.2. Sampling and tagging

The subset chosen for training and validating the AI model was shortened to 2762 patients considering erroneous registrations and missing data (n = 1717 patients), which are frequently found in healthcare databases (Austin et al., 2021; Tayefi et al., 2021). The median was employed to address the latter problem but taking into account possible variations in the dependent variable layers (Guo et al., 2019). In this case, the response variable was the hospitalization admission probability in SRD patients. The hospitalization admission rate was 9 % which denotes a disparity between the positive level (patients not discharged to the HD) and the negative class (patients transferred to the HD). We later implemented random under-sampling on the training sample to stabilize both layers and diminish errors in the AI model (Kang et al., 2020; Gao et al., 2023). In the final sample (n = 1150 patients), this metric was labeled as a binary variable with "1″ denoting a patient transferred to this department; otherwise, "0".

4.3. Significance assessment of potential predictors

Table 2 details the clinical and sociodemographic features of SRD patients involved in the sub-sample. Some interesting insights can be pinpointed from the descriptive statistics and later underpinned by ANOVA tests. In this case, only DAP (Z(DAP_1) = −1.20, *p*-value (DAP_1) = 0.230; Z(DAP_2) = −0.78, *p*-value (DAP_2) = 0.438) and SAP (Z(SAP_1) = −1.41, *p*-value (SAP_1) = 0.157; Z(SAP_2) = −0.45, *p*-value (SAP_2) = 0.342) were found to be non-significant in predicting the hospitalization admission of SRD patients and must be therefore discarded from the AI model. It is good to highlight that various features were concluded to be strongly associated with the outcome (*p*-value = 0). For example, SEX (Z=5.39) evidences that men (n = 292; 73.4 %) are more prone to need hospitalization care compared to women (n = 106; 26.6 %). Regarding LDH, it is noted that the concentration of this enzyme is higher in transferred SRD patients (μ = 764.1 UI/L; SD=413.5 UI/L) than in those who did not need an upstream service (μ = 575.4 UI/L; SD=335.3 UI/L). The same pattern is observed in CPR levels: Hospitalized (μ = 135.94 mg/ml; SD=91.55 mg/ml) vs. non-hospitalized (μ = 83.16 mg/ml; SD=69.45 mg/ml). Also, features indicating the SRD progression showed slight changes from one measure to the other. In the case of HP, the Z-value increased from 2.12 (*p*-value = 0.034) to 2.32 (*p*-value = 0.021), while this metric decreased in OSL from −6.38 (*p*-value = 0.000) to −7.05 (*p*-value = 0.000). All these significant factors (*p*-value < 0.05) are tagged as predictors and can be therefore useful to enhance the accuracy of hospitalization admission decisions greatly. The Mean Decrease of Gini Coefficient (MDGC) calculated for each predictor is shown in Fig. 3. From this figure, we can identify that the features with the highest MDGC are: LDH (87.01), C-RLP (71.09), and DDC (58.01).

Table 2
Profile of SRD patients admitted and not admitted to the hospitalization unit.

Patient features	Admitted to HD	Not Admitted to HD	Z statistic (*p*-value)
Total SRD patients	398 (34.6 %)	752 (65.4 %)	–
Age (A) (years)			−2.01 (0.045)
Mean (SD)	64.9 (15.2)	66.9 (16.4)	
Q1 – Q3	60 to 75	56 to 79	
Sex (S)			
Male	292 (73.4 %)	429 (57.1 %)	5.39 (0.000)
Female	106 (26.6 %)	323 (42.9 %)	
Lactate Dehydrogenase (LDH) (UI/L)			
Mean (SD)	764.1 (413.5)	575.4 (335.3)	8.20 (0.000)
Range	258 to 7,172	223 to 7,172	
C-Reactive Protein Level (C-RPL) (mg/ml)			
Mean (SD)	135.94 (91.55)	83.16 (69.45)	9.16 (0.000)
Range	1 to 564	0 to 560	
D-Dimer Concentration (DDC) (ng/mL)			
Mean (SD)	3098 (9807)	1706 (6974)	2.53 (0.011)
Range	108 to 122,499	113 to 102,622	
Diastolic arterial pressure (DAP) (mm Hg)			
DAP_1 Mean (SD)	76.2 (9.5)	79.6 (46.98)	−1.20 (0.230)
DAP_2 Mean (SD)	76.5 (10.8)	77.8 (30.95)	−0.78 (0.438)
Systolic arterial pressure (SAP) (mm Hg)			
SAP_1 Mean (SD)	130.6 (15.32)	132.37 (22.45)	−1.41 (0.157)
SAP_2 Mean (SD)	131.3 (16.45)	132.47 (22.40)	−0.45 (0.342)
Heart pulse (HP) (# of heartbeats)			
HP_1 Mean (SD)	91.8 (14.3)	89.67 (16.97)	2.12 (0.034)
HP_2 Mean (SD)	92.04 (15.3)	89.67 (16.96)	2.32 (0.021)
Oxygen saturation level (OSL) (%)			
OSL_1 Mean (SD)	90.02 (8.58)	93.17 (5.95)	−6.38 (0.000)
OSL_2 Mean (SD)	89.46 (9.60)	93.15 (5.63)	−7.05 (0.000)

4.4. Training, testing, and validating the AI model

The final sample (n = 1150 patients) was randomly split into training (n = 810 patients) and test (n = 340 patients) cohorts using an approximate 70/30 partition. We used the Rstudio software (v. 4.1.2) to deploy the RF algorithm and derive the predictive model. Specifically, we utilized the following libraries: epiR (v.2.0.44) (Ortiz-Barrios et al., 2023), Caret (v.6.0–90) (Ning et al., 2023), randomForest (v.4.7–1), and ROCR (v.1.0–11). Also, we executed a 10-fold cross-validation to calibrate the hyperparameters.

The model performance was appraised considering different indicators (Table 3). First, McNemar's test *p*-value (0.1859) was greater than the significance level (0.05) which underpins the homogeneity hypothesis between the proportions of misclassified patients corresponding to the two layers. On a different tack, most indicators overpass 90 %, which denotes excellent distinction by the AI algorithm when pinpointing SRD patients with/without hospitalization care needs. This is additionally validated by the Receiving Operator Characteristic (ROC) curve plot (Fig. 4), which presents an AUC of 96.28 % (95 %CI, 90.48 % − 100 %). Similarly, the specificity (85.0 %, 95 %CI (77.33 % − 90.86 %)) indicates that out of 100 SRD patients without hospitalization needs, the AI model will predict between 77 and 91 suitably. Meanwhile, the sensitivity (95.45 %, 95 %CI (91.79 % − 97.79 %) denotes that out of 100 SRD patients requiring hospitalization care, the AI model will correctly forecast between 91 and 98. On a different note, the positive predictive value (92.11 %, 95 %CI (87.80 % − 95.25 %) uncovers that between 87 and 96 of every 100 respiratory-infected patients with hospitalization entrance forecast, will be finally transferred to this department. In the interim, the negative predictive value (91.07 %, 95 % CI (84.19 % − 95.63 %) exposes that between 84 and 96 out of 100 SRD patients with non-hospitalization transfer prediction will not be lastly moved to this area.

4.5. The simulation model: From construction to validation

The integration between AI and DES occurs when the predicted hospitalization probabilities are inserted as a vector into the simulation model. Thereby, it is possible to: i) Measure the expected hospitalization bed waiting time that these patients may experience under the current operational conditions, ii) Define if the existing bed installed capacity is sufficient to address the forecasted hospitalization admissions, and iii) Determine how the HD can be reorganized or what other alternatives may be explored to lessen the hospitalization bed waiting times. The following sub-sections will outline how these aims were coped with and what strategies were found to be effective for ensuring the timely seizure of hospitalization beds by SRD patients.

4.5.1. Diagramming the simulation model

The simulation model must provide a reliable imitation of the real-

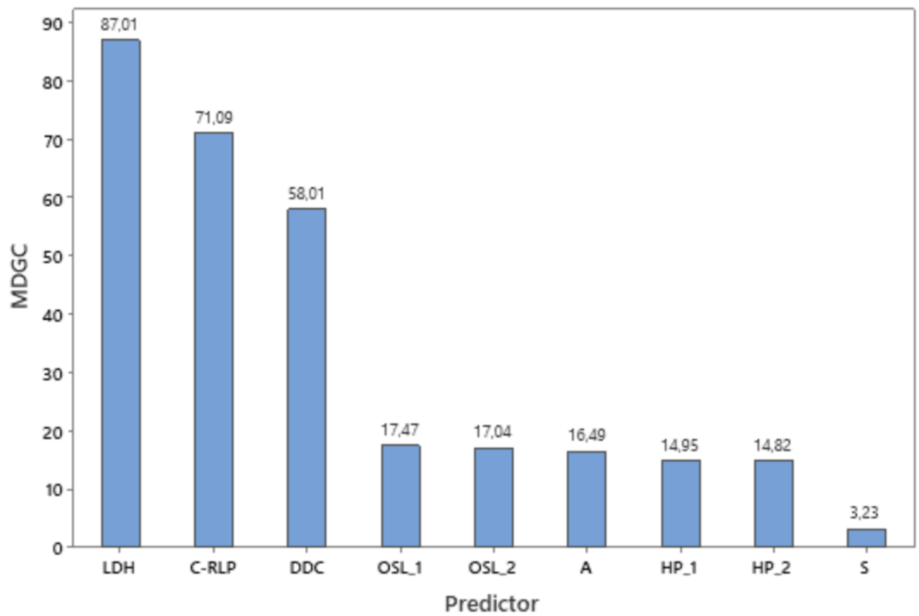

Fig. 3. MDGC for hospitalization admission predictors in SRD patients.

Table 3
Performance indicators of the RF model.

Performance indicator	Score (95 % CI)
Sensitivity	95.45 % (91.79 % – 97.79 %)
Specificity	85.00 % (77.33 % – 90.86 %)
Accuracy	91.44 % (88–06 % – 94.11 %)
Positive predictive value	92.11 % (87.80 % – 95.25 %)
Negative predictive value	91.07 % (84.19 % – 95.63 %)
Area Under Curve (AUC)	96.28 % (90.48 % – 100 %)
Mcnemar test p-value	0.1859

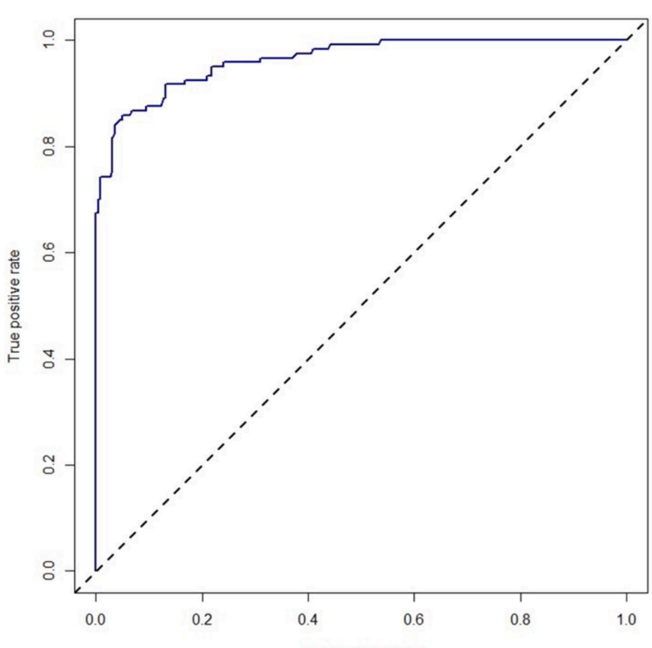

Fig. 4. ROC curve for the test cohort in the predictive hospitalization admission model.

world interface between the emergency department, hospitalization unit, and intensive care service. To ensure this, Gemba walks were performed to analyze the patient's journey during a real SRD. The insights collected in this activity were later complemented with the information stored in the Quality Management System (QMS) and the comments of medical/administrative staff involved in the services mentioned above. The main outcome is a flow diagram itemizing the SRD patient experience from the emergency services to intensive care (Fig. 5).

4.5.2. *Input data analysis*

The process diagramming also enabled us to pinpoint five main process variables: Time between admissions of SRD patients, Triage time, ED Length of Stay (ED-LoS), HD-LoS, and ICU-LoS. Following this, auto-correlation tests were conducted to assess the randomness of these metrics at a significance level of 1 %. The results supported the independence of the variables as the absolute values of T limits do not overpass 2 (Table 4). As the next step, homogeneity analyses were conducted using the Welch's test (error level = 0.01) (Table 5). The outcomes confirmed the presence of diverse admission patterns considering the weekday and time range (F=4.85; *p*-value = 0), different triage times based on the classification categories (1–2, 3–5) (F=15.95; *p*-value = 0.000), and diverse UCI-LoS based on the health status of SRD patients (poor, moderate). A probability distribution is necessary to describe each subgroup. The rest of the process variables were homogeneous (*p*-value > 0.05) and one stochastic model was sufficient to denote their behavior fully. Ultimately, KS tests (α = 0.01) were conducted to elucidate the probability expression better representing each process variable (Table 6).

4.5.3. *Creation and validation of the DES model*

Once input data pre-processing was completed, Arena® 16.10.00 software was employed to design a useful virtual imitation of the healthcare pathway followed by the SRD patients. The patient pathway outlined in Fig. 2 and the stochastic models derived from the KS tests were inserted in this simulation. The replication period was determined to be 15 days (24 h/day) as ED, HD, and ICU function at a 24/7 rhythm. In addition, a warm-up cycle of 120 days was defined as necessary to reach the system stability (blocking likelihood = 0). After this, the virtual healthcare system was run ten times to calculate the final volume of iterations that would be deemed during validation. The ED waiting

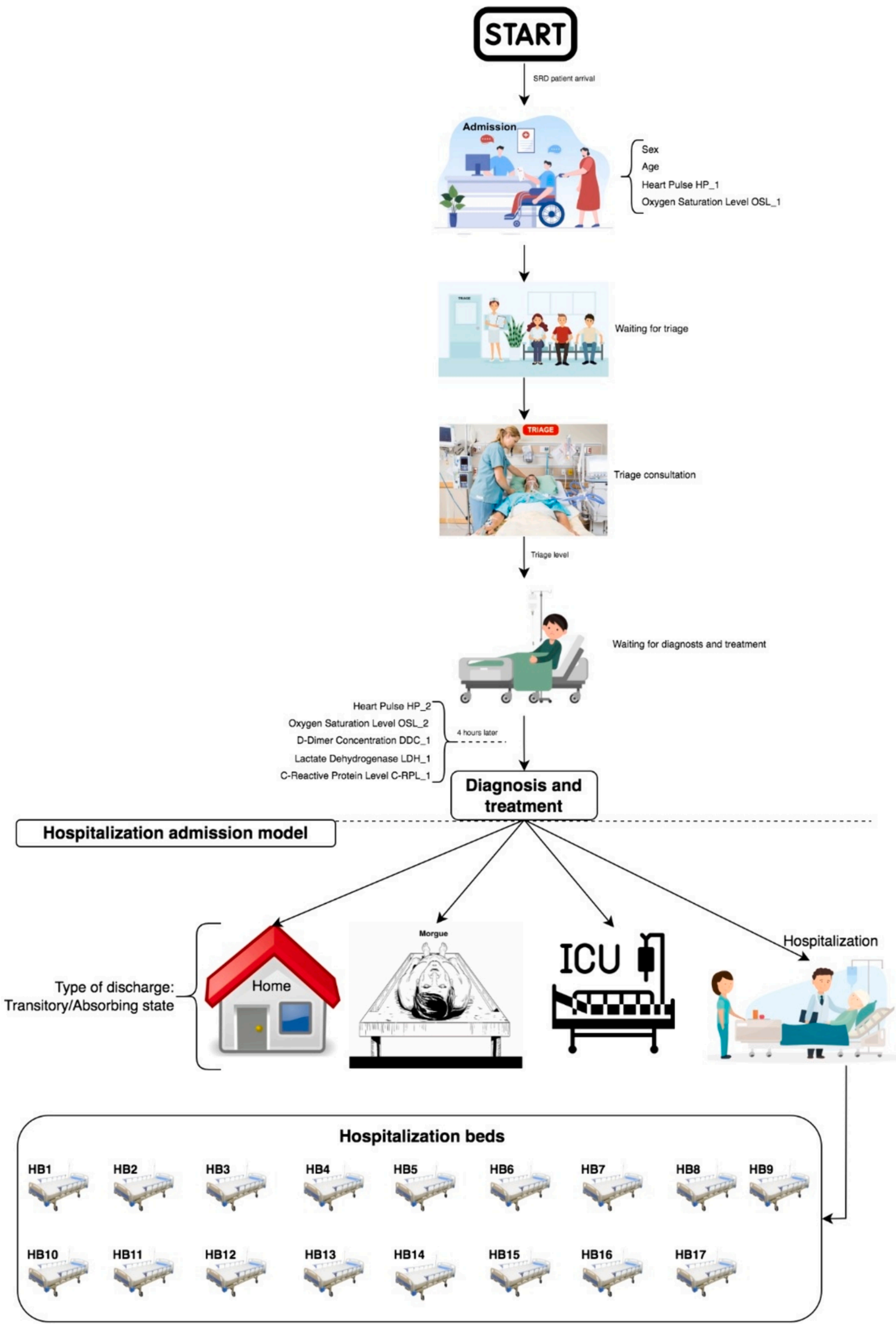

Fig. 5. SRD patient pathway within the medical center.

times for III-V triage categories and the hospitalization bed waiting time were derived in each run. In this model, 337 iterations were needed to represent the present variability of the real healthcare configuration fully. Ultimately, a 1-sample *t*-test ($\alpha = 0.01$) was performed to compare the real and simulated hospitalization bed waiting times (Ho: $\eta = 29$ h || Ha: $\mu \neq 29$ h; t = 0.10; *p*-value = 0.921). The same analysis was undertaken for the ED waiting time in III-V triage categories (Ho: $\mu = 5$ h || Ha: $\mu \neq 5$ h; t = 0.25; *p*-value = 0.810). The validation outcomes uncovered that the virtual is statistically equivalent to the real-world system and can hence be employed for operational analysis and pre-testing of process improvements. Fig. 6 reports the capability analysis regarding the current hospitalization bed waiting time experienced by SRD patients. This metric's Upper Specification Limit (USL) is 5 h/patient. The results indicate that the average waiting time is 29.204 h with a standard

Table 4
Randomness test results.

Process variable	T [Min; Max]
Time between admissions of SRD patients	[−0.0200361; −0.0207823]
Triage time	[−1.43; 1.76]
ED Length of Stay (ED-LoS)	[−1.86; 1.74]
HD-LoS	[−1.91; 1.74]
ICU-LoS	[−1.92; 1.97]

Table 5
Homogeneity test outcomes.

Process variable	F-value (*p*-value)	Conclusion
Time between arrivals of SRD patients	4.85 (0.000)	Heterogeneous per time slot-day
Triage time	15.95 (0.000)	Heterogeneous per triage category
ED Length of Stay (ED-LoS)	2.44 (0.046)	Homogeneous
HD-LoS	0.19 (0.665)	Homogeneous
ICU-LoS	24.80 (0.000)	Heterogeneous per health status

deviation of 38.058 h (95 % CI 25.126 – 33.282 h). Furthermore, 73.76 % of the data is out of specification, resulting in a low and negative capability index (Z=-0.64). The bimodal histogram is bimodal, comprising one behavior below the USL, explained by the immediate availability of beds (Hospitalization bed waiting time = 0), and the other above this reference, indicating delay. In summary, out of 1,000,000 of patients requiring hospitalization services, 737,608 will have to wait for more than 5 h before bed allocation.

4.5.4. *Pretesting improvement proposals*

Some seasonal respiratory diseases provoke a fast worsening of health conditions, which entails a rapid provision of healthcare in upstream services when required (Lin et al., 2020). In this case, it is indispensable to forecast how probable a patient is to be transferred to the hospitalization unit and then simulate how this department can be reconfigured to lessen prolonged bed waiting times as evidenced in the previous subsection. In this regard, we worked closely with the medical and administrative staff to design two strategies to reduce the current gap between the expected and current hospitalization bed waiting time. The simulated model was utilized to pretest these proposals to verify their operational goodness before further implementation in the wild. Thereby, we can ensure that scarce hospital's financial resources are correctly allocated. The suggested strategies were: (i) enable two additional hospitalization rooms with one bed each, implement physician rounding in the early morning, and apply lean techniques to reduce delays due to patient exit authorization, patient transport, and insurance documentation, (ii) enable two additional hospitalization rooms with one bed each and transform two triage rooms into hospitalization units. The remedies have been appraised in terms of hospitalization bed waiting time contrasted with a non-intervention (NIS) context (Fig. 7). The interval graph evidences that remedy ii is the best solution with a hospitalization bed waiting time oscillating between 17.14 and 25.35 h, which is lower compared to the time range currently offered by the medical center (25.126 – 33.282 h).

We additionally conducted a Mann-Whitney test to confirm the previous findings. The outcomes revealed that Proposal (ii) would trigger a significant reduction in the waiting time for hospitalization beds (W=105,618, *p*-value = 0.004) if implemented. Meanwhile, Proposal (i) would not cause this same effect (W=69,404, *p*-value = 0.143) due to its instability, as observed in Fig. 6.

Table 6
Probability expressions of process variables inserted into the simulation model.

Process variable/Pipeline		Probability model	KS (*p*-value)
Time between admissions of SRD patients	Monday – TS1	GAMM(2.03, 16.3) hours	0.13 (>0.15)
	Monday – TS2	−0.02 + LOGN(0.44, 0.95) hours	0.06 (>0.15)
	Monday – TS3	−0.02 + LOGN(0.31, 0.48) hours	0.06 (0.24)
	Tuesday – TS1	WEIB(1.52, 19.41) hours	0.16 (>0.15)
	Tuesday – TS2	−0.02 + LOGN(0.53, 1.01) hours	0.03 (>0.15)
	Tuesday – TS3	−0.02 + LOGN(0.30, 0.44) hours	0.06 (>0.15)
	Wednesday – TS1	WEIB(1.49, 14.01) hours	0.07 (>0.15)
	Wednesday – TS2	−0.02 + WEIB(0.54, 18.36) hours	0.10 (0.08)
	Wednesday – TS3	−0.02 + LOGN(0.32, 0.43) hours	0.07 (>0.15)
	Thursday – TS1	WEIB(1.18, 19.03) hours	0.11 (>0.15)
	Thursday – TS2	−0.02 + LOGN(0.71, 1.54) hours	0.06 (>0.15)
	Thursday – TS3	−0.02 + LOGN(0.30, 0.47) hours	0.07 (>0.15)
	Friday – TS1	WEIB(1.16, 16.5) hours	0.10 (>0.15)
	Friday – TS2	−0.02 + LOGN(0.58, 1.19) hours	0.04 (>0.15)
	Friday – TS3	−0.02 + LOGN(0.32, 0.53) hours	0.09 (0.08)
	Saturday – TS1	−0.02 + GAMM(1.24, 20.06) hours	0.09 (>0.15)
	Saturday – TS2	−0.02 + LOGN(0.68, 1.35) hours	0.09 (>0.15)
	Saturday – TS3	−0.02 + LOGN(0.39, 0.61) hours	0.08 (>0.15)
	Sunday – TS1	−0.02 + GAMM(2.15, 13) hours	0.20 (0.07)
	Sunday – TS2	−0.02 + LOGN(0.60, 1.16) hours	0.06 (>0.15)
	Sunday – TS3	−0.02 + GAMM(0.43, 22.56) hours	0.11 (0.07)
Triage time	1–2	UNIF(3.5, 11.5) minutes	0.06 (0.24)
	3–5	TRIA(3.5, 4, 11.5) minutes	0.03 (0.17)
ED-LoS		(10 + 35*BETA(0.936, 1.01))/7.5 h	0.02 (>0.15)
HD-LoS		9 + 13*BETA(1.14, 1.18) days	0.03 (>0.15)
ICU-LOS	Moderate	42*BETA(0.495, 1.65) days	0.05 (>0.15)
	Poor	WEIB(17.1, 0.969) days	0.03 (>0.15)

TS1: Time slot from 08:00 to 16:00; TS2: Time slot from 16:00 to 00:00; TS3: Time slot from 00:00 to 08:00.

5. Discussion

5.1. *Discussion on the results of the AI model*

The AI model developed in this study focuses on analyzing demographic and clinical factors influencing the decision to admit SRD patients to HDs. By leveraging the random forest algorithm, the model predicts the likelihood of SRD patients being directed to HDs and feeds into a DES model designed to optimize bed capacity planning in these units, thereby reducing bed waiting times. Notably, variables such as Sex (Gershon et al., 2017; Sun et al., 2018; Yanamala et al., 2021; De Freitas et al., 2022), Age (Gershon et al., 2017; Sun et al., 2018; Ma et al., 2020; Yanamala et al., 2021; De Freitas et al., 2022), Diastolic Arterial

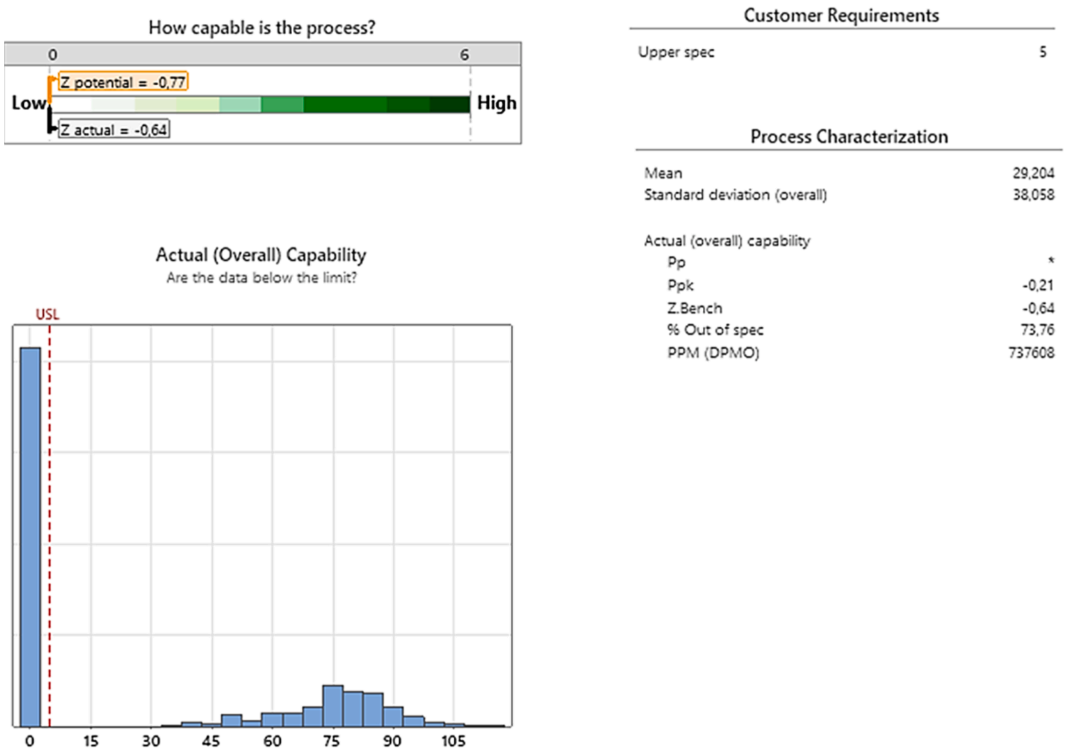

Fig. 6. Capability analysis on the current hospitalization bed waiting time.

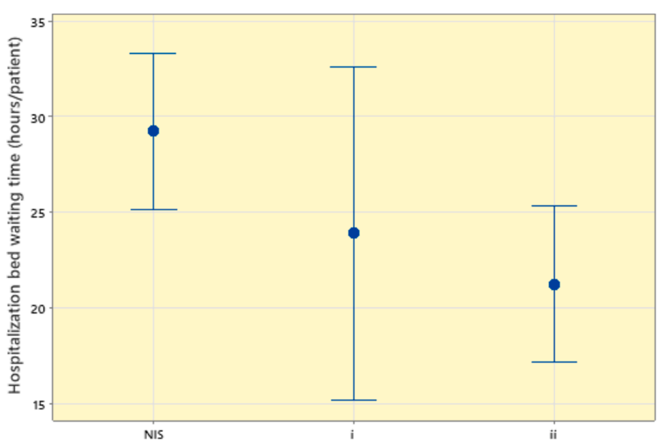

Fig. 7. Hospitalization bed waiting times in NIS, i, and ii.

Pressure (Yanamala et al., 2021; De Freitas et al., 2022), Systolic Arterial Pressure (Yanamala et al., 2021; De Freitas et al., 2022), Heart Pulse (Yanamala et al., 2021; De Freitas et al., 2022), Oxygen Saturation Level (Yanamala et al., 2021; De Freitas et al., 2022), D-Dimer Concentration (Ortiz-Barrios et al., 2023), Lactate Dehydrogenase (Ortiz-Barrios et al., 2023), and C-Reactive Protein Level (Ortiz-Barrios et al., 2023) were included in the RF model. The study found that while Diastolic and Systolic Arterial Pressure did not significantly predict hospitalization decisions for SRD patients, other variables proved to be statistically significant predictors at the specified confidence level. An important point here is that the majority of studies in the literature on predicting the admission of SRD patients to HDs determine that this decision is based on weather-based variables (such as humidity, humidity, pressure, and precipitation), and air pollution-based variables (nitrogen oxides, sulfur dioxide, carbon monoxide, ozone, particulate matter of median aerometric diameter etc.) depends on how it is affected. The literature on this subject is abundant and full (Shakerkhatibi et al., 2015; Navares et al., 2018; Becerra et al., 2020; Xavier et al., 2022; Yang et al., 2023). Our study fills an important gap since it addresses an important issue outside this field where many studies have been conducted.

Furthermore, the study employed the MDGC criterion to assess the importance of predictive variables for hospital admission decisions in SRD patients. Lactate Dehydrogenase, C-Reactive Protein Level, and D-Dimer Concentration were identified as the most critical predictors, followed by Oxygen Saturation Level measured at the triage station and four hours after admission. These findings align with similar studies that highlight specific biomarkers as crucial indicators for hospitalization in respiratory diseases. The literature De Freitas et al. (2022) study on predicting hospitalization in patients with respiratory symptoms during the covid-19 pandemic, it was concluded that lower saturation and higher respiratory rate, higher heart rate and lower blood pressure levels, age, and male gender require hospitalization more often.

The model's performance was evaluated using the AUC-ROC value, which measures its ability to distinguish between admitted and non-admitted SRD patients at HDs. The study reported an AUC value of 96.28 %, indicating excellent discriminatory power compared to similar studies in the literature, which typically report around 92 %. This superior performance underscores the effectiveness of the RF algorithm in predicting hospital admissions based on clinical data.

5.2. Discussion on the results of the DES model

In addition to the AI model, this study integrated a DES model to optimize the allocation and management of SRD patients in HDs. Unlike traditional DES applications, which focus primarily on operational dynamics, this study extends its scope to include detailed probabilistic modeling of clinical variables and their impacts on patient flow and resource utilization. Hosseini Shokouh et al. (2022) optimized the hospitalization unit, as well as the hospitalization units' doctors, by DES, Artificial Neural Network (ANN) algorithm, and Genetic Algorithm (GA). Although various AI-based algorithms (whether these are a demand forecasting model given as input to DES or models that provide decision support for optimization purposes using the output of DES) are integrated into DES to improve the effective management of critical resources of emergency services such as beds are found in the literature (Zeinali et al., 2015; Nas and Koyuncu, 2019; Harper and Mustafee, 2019; Atalan et al., 2022; Ordu et al., 2023), it is clear that the problem at the focus of the current study is unique.

Integrating AI and DES models in healthcare resource management are increasingly recognized for its potential to enhance decision-making and operational efficiency. By simulating patient flows and bed allocations in real-time scenarios, hospitals can anticipate demand surges and optimize resource allocation accordingly. Previous studies (Zeinali et al., 2015; Nas and Koyuncu, 2019; Harper and Mustafee, 2019; Atalan et al., 2022; Ordu et al., 2023) have shown that integrating AI-driven demand forecasting with DES modeling can lead to significant improvements in patient outcomes and operational costs.

Furthermore, the study underscores the importance of fitting probability distributions and conducting goodness-of-fit tests within the DES modeling process. These methodological refinements ensure the accuracy and reliability of simulations, enabling healthcare providers to make informed decisions based on robust data analysis.

5.3. *Managerial and policy implications*

The implications of this study extend to healthcare managers and policymakers tasked with optimizing hospital operations during respiratory disease outbreaks. SRDs pose significant challenges to hospital emergency departments (EDs) and HDs (Boyle et al., 2022; Fava et al., 2022), leading to overcrowding, prolonged patient wait times, and resource shortages (Gul and Celik, 2020). Such problems occurring in EDs also disrupt the settings of HDs and cause excessive waiting in these units, especially for beds (Devapriya et al., 2015). By combining AI predictions with DES simulations, healthcare leaders can proactively manage patient admissions and bed capacities, thereby improving overall service delivery and patient outcomes.

AI-based models contribute to solving the problems mentioned above with inferences based on large patient data in hospital EDs and HDs. Making these inferences correctly depends on whether the variables in the data set are meaningful for the AI-based model. Here, it is important to develop a model whose explanatory predictors are statistically significant and perform superior to equivalent models in the literature rather than a model that includes all variables.

Decision or policymakers within the hospital environments use the prediction results obtained using the RF algorithm in terms of the probability of admission to HD and determine which configurations will be effective in reducing HD bed waiting times through the DES model. Thus, a solution will be provided to the problem of bed adequacy, which is the weakest link for the hospital in cases where demand reaches high levels during the respiratory disease season. In this way, hospital decision-makers can understand the impact of different configurations on bed waiting times and implement the best one before the situation becomes much more dire.

6. Conclusions

Emergency care processes related to seasonal respiratory diseases traditionally present problems for emergency services. Among the most common are: the sizing of available resources (healthcare personnel, care spaces, beds, ventilators, etc.) and the medical decision to be taken with the patient (diagnosis, treatment, and hospitalization). Both types of problems are directly related and can have a mutual influence. In this sense, the interaction of multiple variables (patient waiting time, patient evolution according to severity, typology and clinical parameters, sociodemographic and environmental factors, etc.) makes designing and managing emergency services complex. Therefore, this paper presents a hybrid approach combining Artificial Intelligence and Discrete-Event Simulation to shorten the bed waiting times in hospitalization departments. A case study of a European hospital group was used to validate the suggested framework.

Integrating both techniques (AI-DES) enhances the benefits already provided by each of them individually, considering that dealing with complex problems in unpredictable situations makes it necessary to seek hybrid approaches by combining techniques that have demonstrated their potential to deal with a multitude of problems of varied etiology. The proposed approach allows decision-makers to anticipatedly define if an SRD patient will be transferred to the hospitalization unit in the next few hours with the aid of AI models. The DES takes these predictions to evaluate if this unit will provide timely care to expected demand and pretest potential interventions that may be implemented in case of a significant gap between the bed waiting time reference and the current operational performance.

High-quality datasets are needed for feeding and training AI-DES models. Consequently, healthcare organizations will be able to identify the hospitalization drivers based on models using significant variables gathered from the downstream stages of the healthcare journey. In this sense, an analysis has been performed to enumerate the hospitalization admission features in SRD patients, which, in addition, have been contrasted with the medical staff.

The subset chosen for training and validating the AI model was obtained from a raw dataset supplied by the hospital group, to which a filtering process was applied. The final sample ($n = 1150$ patients) was randomly divided into training and test cohorts, and Rstudio software (v. 4.1.2) was used to implement the Random Forest algorithm to derive the predictive model. For the model evaluation, several performance indicators were used, obtaining, for most of them, values exceeding 90 %, which denotes an excellent distinction by the AI algorithm in identifying patients with SRD with or without hospital care needs.

The results obtained by the AI model show some interesting considerations. Firstly, Diastolic arterial pressure and Systolic Arterial Pressure are not statistically significant predictors in the hospitalization decision of SRD patients to HDs; and secondly, although the remaining variables contain statistically varying results, it is seen that they are important in the prediction and confirm to be good hospitalization admission predictors at the given confidence level. One of the most relevant conclusions is that using the RF algorithm in the developed model makes an excellent distinction between SRD patients who have an admitted/ not admitted decision at the HDs. This is very relevant because it allows for predicting the possibility of respiratory worsening based on sociodemographic and clinical patient data.

On the other hand, the simulation model construction is particularly delicate, as it must imitate as closely as possible the real-world interface between the emergency department, the inpatient unit, and the intensive care department. Therefore, it was considered appropriate to perform Gemba walks to analyze the patient's journey during a real SRD and later complement their findings with the information stored in the Quality Management System (QMS) and the comments of medical/ administrative staff. This process allowed us to pinpoint five main process variables: Time between admissions of SRD patients, Triage time, ED Length of Stay (ED-LoS), HD-LoS, and ICU-LoS. After the tests to assess the randomness of these metrics and the homogeneity analyses, diverse admission patterns were confirmed, considering the weekday and time range, different triage times based on the classification categories, and diverse ICU-LoS based on the health status of SRD patients. In addition, it was necessary to determine the probability expression that best represents each process variable by performing KS tests. Arena® 16.10.00 software was used for the simulation, using the patient pathway and the stochastic models derived from the KS tests as main input variables.

The results demonstrate the robustness of the model created and can, therefore, be used for performance diagnosis and further pre-testing of operational improvement interventions. Despite this, our work holds some limitations: i) Any deviations in real-world settings (e.g., sudden changes in staff availability and the inclusion of new technology) could impact the model's accuracy. More dynamic modeling approaches could be explored to explain such uncertainties, ii) It is important to mention that the database used in this study contained only sociodemographic features, vital signs, laboratory test results, and prescribed treatments collected in the emergency department. In this regard, another limitation is the non-incorporation of external factors such as weather conditions or air pollution that could influence respiratory disease patterns in patients (Navarro Romero et al., 2020), iii) The improvement scenarios pretested in the virtual model are restricted to bed and room arrangements. Investigating a wider range of interventions, such as staffing variations or policy changes, could offer more comprehensive solutions to shortening bed waiting times, and iv) The case study entails a European hospital group; therefore, the outcomes may not be directly applicable to hospitals in different regions with fluctuating healthcare systems and resource constrictions. Future research should test the model in diverse settings to improve its generalizability.

Future work can delve into the use of new AI models based on improved algorithms that contribute to analyzing the decision-making process in hospital centers under different perspectives and factors, knowing that these elements condition both the sample of data used in the AI model as well as in the development of the simulation model. In addition, the DES model can be extended to some support services, including X-ray images and clinical lab, enabling the design of more profound approaches analyzing the influence of these processes on the overall hospital stay. Also, it is intended to incorporate scheduling models optimizing the use of doctors and nurses focusing on minimizing delays in treatment and diagnosis. Future efforts may additionally consider using oversampling techniques and synthetic data generation to tackle the data imbalance problem and consequently improve the model's robustness.

CRediT authorship contribution statement

Miguel Ortiz-Barrios: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Alessio Ishizaka:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Maria Barbati:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Sebastián Arias-Fonseca:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jehangir Khan:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Muhammet Gul:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Melih Yücesan:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Juan-Jose Alfaro-Saíz:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Investigation, Formal analysis, Conceptualization. **Armando Pérez-Aguilar:** Writing – review & editing, Writing – original draft, Visualization, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the European Union Next Generation EU under the Margarita Salas grant launched by Universitat Politècnica de València (Recovery, Transformation, and Resilience Plan) and Ministerio de Ciencia, Innovación y Universidades (Program for Retraining of the Spanish University System 2021-2023).

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